lecture 2

Smart Devices

LECTURE 3

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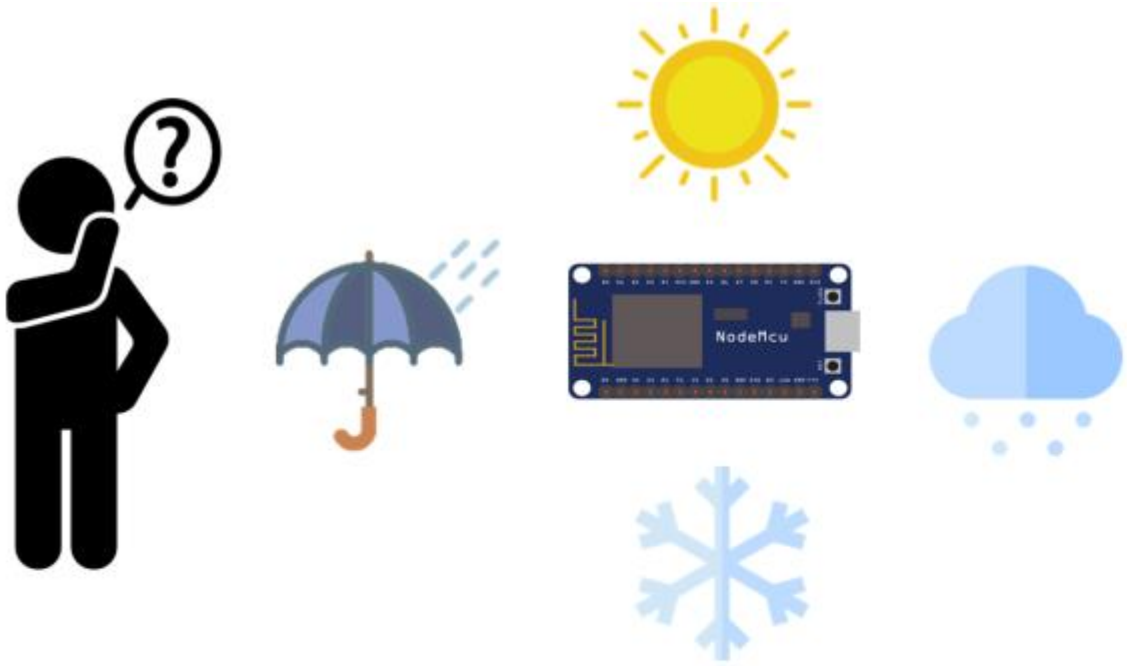
Sensor technologies and IOT Protocols for smart devices

IoT Applications

1. **Smart Umbrella**
   - An umbrella that provides information about the likelihood of rain so that users can decide whether to take the umbrella with them as they leave their homes.
   - The umbrella has a handle that would **illuminate** when snow or rain is forecasted.
   - Using existing **Wi-Fi** technology to pull information about the weather from the Internet.



IoT Applications

2. **Quirky Egg Minder**
   - When your egg supply gets low, this IoT application will send info directly to your phone to remind you to buy more eggs.
3. **WELT**
   - WELT cares for the user's overall health by measuring waist size, steps, sitting time, and overeating habits with the sensing technology.

IoT Applications

5. **Smart Toaster**
    - You can use your smartphone to set the darkness of your toast.
    - If a friend has the same toaster, you can send them a picture on toast.

IoT Applications

6. **Amazon Echo Look**
   - Amazon's Echo Look will judge how you look.
   - It compares two outfits and rates which one is better.

IoT Applications

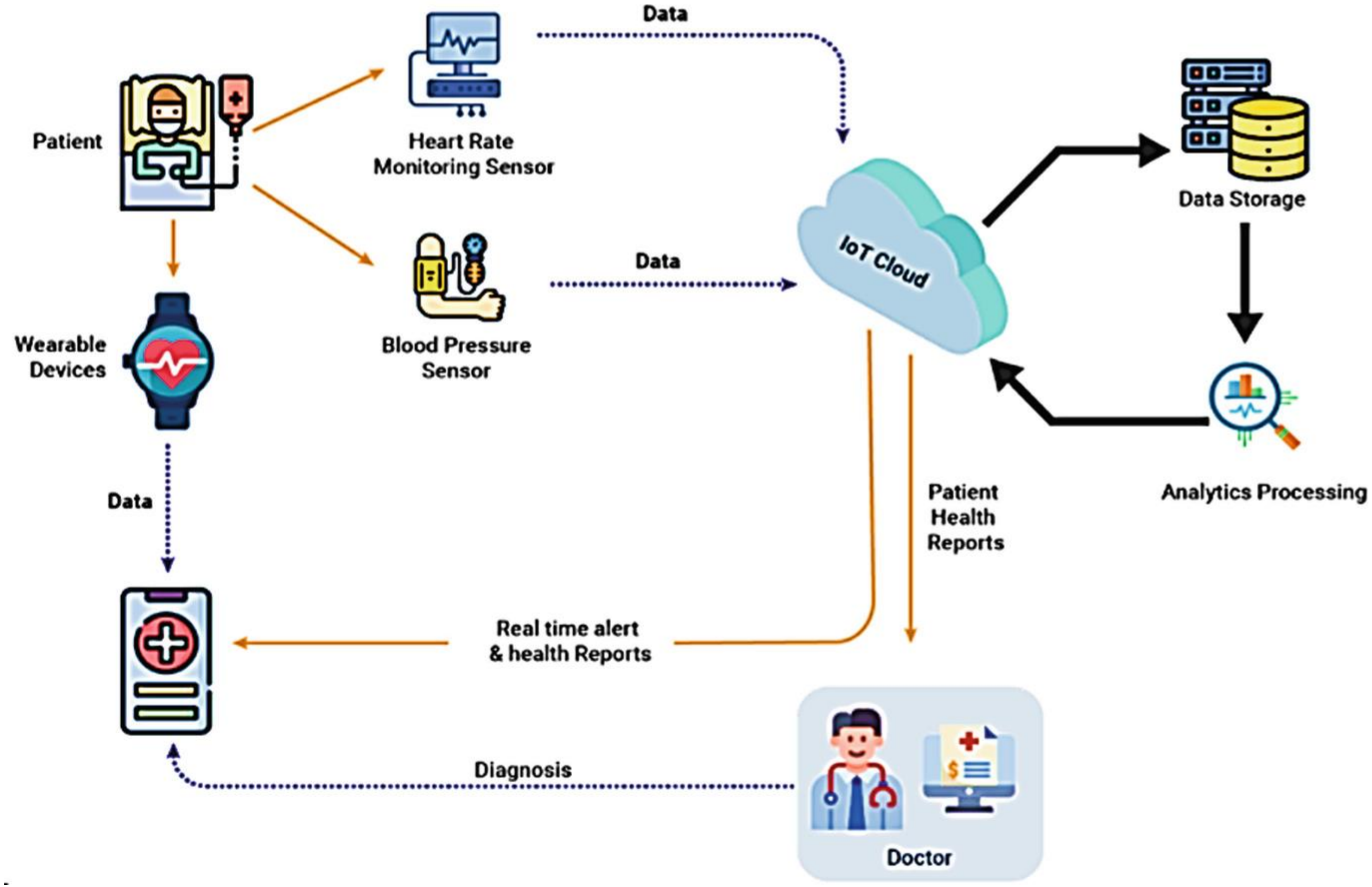
7. **HapiFork**
   - The HapiFork is a Bluetooth-enabled “smart fork” that vibrates when it senses you’re eating too fast.

IoT Applications

8. **Smart Shoes**
   - Smart shoes allow users to change the color of the shoe with one tap on their smartphone.
   - Blind users sensors

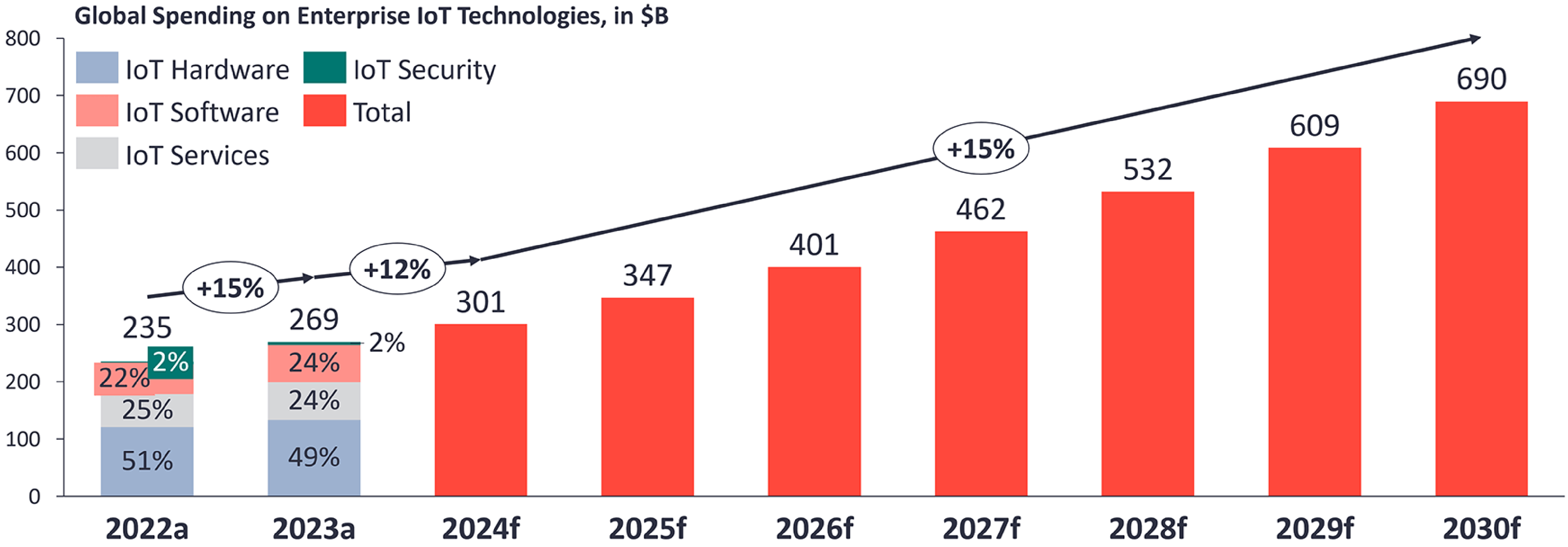
IoT Applications

9. **Healthcare**



The enterprise IoT market by technology 2023–2030

Data as of June 2024



Note: IoT Analytics defines IoT as a network of internet-enabled physical objects. Objects that become internet-enabled (IoT devices) typically interact via embedded systems, some form of network communication, or a combination of edge and cloud computing. The data from IoT-connected devices is often used to create novel end-user applications. Connected personal computers, tablets, and smartphones are not considered IoT, although these may be part of the solution setup. Devices connected via extremely simple connectivity methods, such as radio frequency identification or quick response codes, are not considered IoT devices. Since the last update in 2023 our definition of the enterprise IoT tech stack slightly changed.
a: Actuals, f: Forecast
Source: IoT Analytics Research 2024 – Global IoT Enterprise Spending Dashboard (Q2/2024 update). We welcome republishing of images but ask for source citation with a link to the original post or company website.

Enterprise IoT Market Overview

How Big Is the IoT?

- By **2019**, 10 billion devices were connected to the IoT.
- By **2025**, predictions estimate that there will be nearly 40 billion.

Statistics & Insights

- Market Size: Reached $269 billion in 2023, with 15% year-over-year growth.
- Growth Trend: Expected to slow to 12% in 2024 before accelerating again in 2025.

- Leading Regions: China, India, and the US will drive spending growth.
- Top Industries: Automotive and process manufacturing will lead IoT adoption.

- Software vs. Hardware: Enterprise IoT software spending will grow faster than overall IoT spending.

For more details, https://iot-analytics.com/iot-market-size/

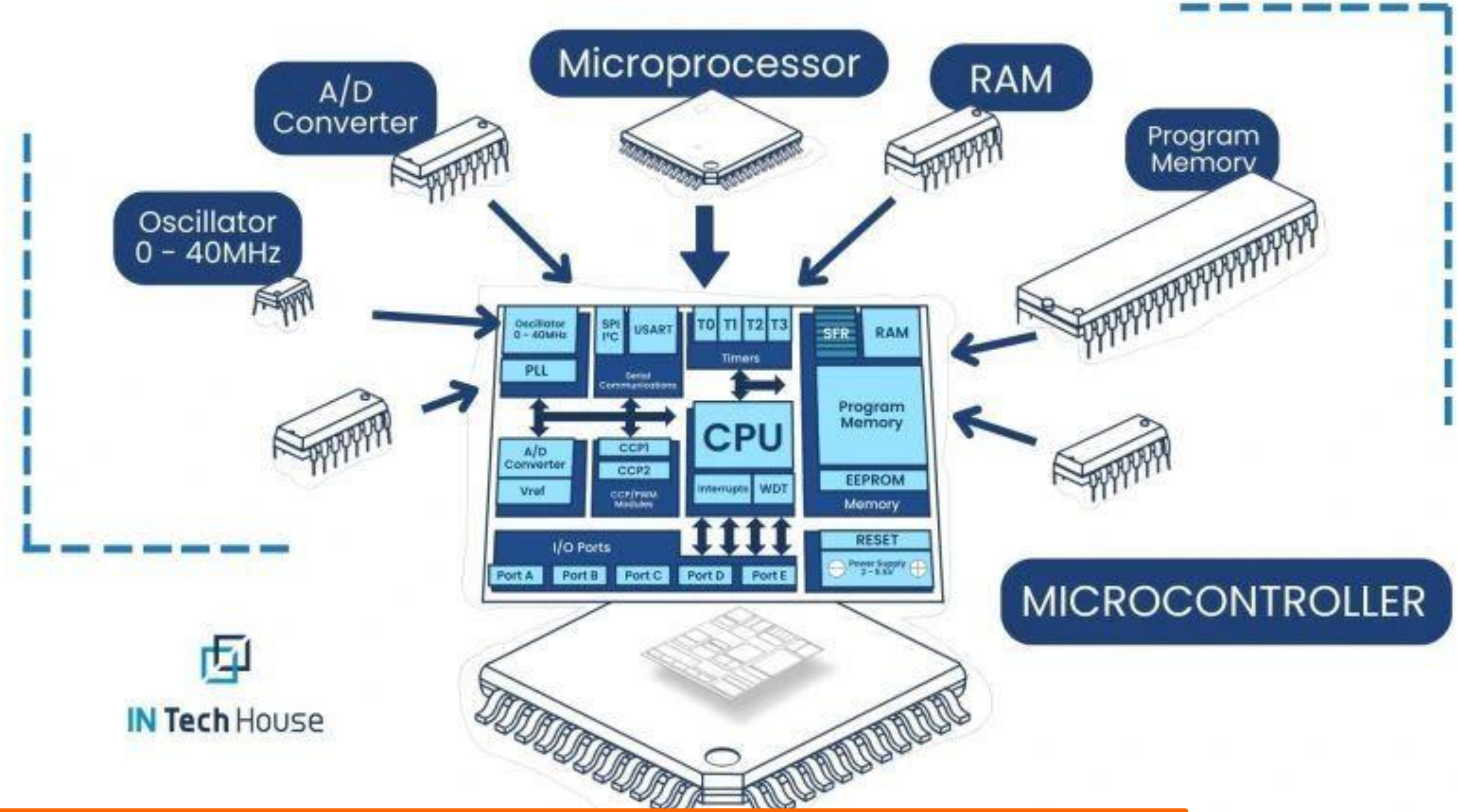
Introduction to Microcontrollers

- **What is Microcontroller!**
  - Microcontrollers (MCUs) are compact integrated circuits that control devices.
  - They consist of a processor, memory, and input/output peripherals.
  - Commonly used in embedded systems, IoT devices, and automation.

Key Characteristics

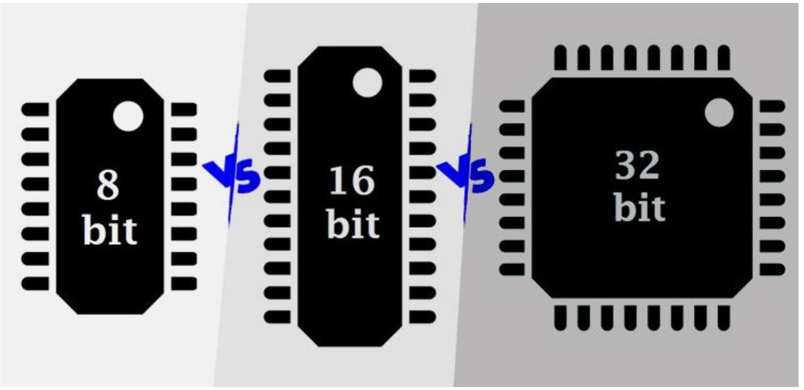
- **Embedded focus single purpose**
- **Compact and efficient all in chip**
- **Real time control**

- **Home app. IOT devices**



For more details, https://www.linkedin.com/posts/anis-hassen_how-to-choose-the-right-mcu-for-your-activity-7280956960135401472-8ynb/

Microcontroller Architecture



- **Microcontroller Architecture**
  - It refers to the internal organization and design of a microcontroller's core components.
  - It defines how the microcontroller processes data, accesses memory, manages peripherals, and executes instructions.

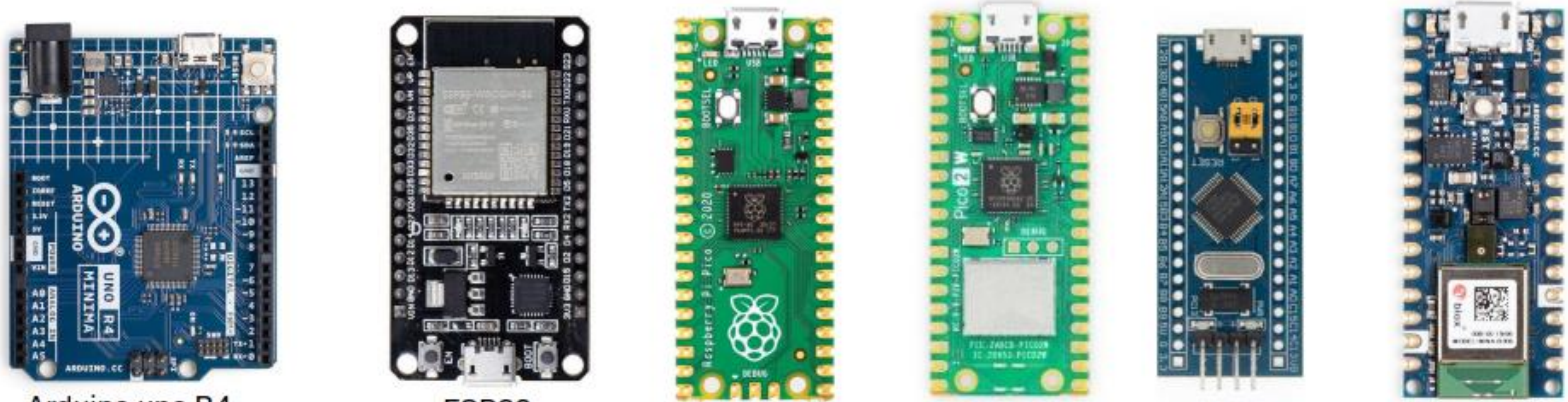
Feature	8-bit Microcontrollers	16-bit Microcontrollers	32-bit Microcontrollers
Data Processing	8-bit words	16-bit words	32-bit words
Performance	Lower	Moderate	High
Memory	Limited	More	Much More
Peripherals	Basic	More Features	Extensive & Advanced
Power Consumption	Lowest	Moderate	Higher
Complexity	Simplest	More Complex	Most Complex
Cost	Lowest	Moderate	Highest
Applications	Simple control, toys, basic devices	Motor control, industrial automation	Complex systems, image processing, embedded OS

For more details, https://components101.com/article/difference-between-8-bit-16-bit-and-32-bit-microcontrollers

Popular Microcontrollers

Arduino uno R3 | Arduino Mega | Arduino Nano | STM08

8-bit



Arduino uno R4 | ESP32 | Pico | PicoW | STM32 | Nano 33

32-bit

Popular Microcontrollers

- **8-bit Microcontroller Units (MCUs)**
  1) **Arduino Uno R3**: A versatile**, beginner-friendly board** with a good balance of features for **general-purpose projects. (2 KB RAM**)
  2) **Arduino Mega**: **A larger, more powerful** board with **more pins and memory** for **complex projects. (8 KB RAM)**
  3) **Arduino Nano**: A compact version of the Uno, ideal **for space-constrained applications. (2 KB RAM)**
  4) **STM8**: **A low-cost**, **low-powe**r 8-bit microcontroller suitable for **simple embedded applications. (1 KB – 6 KB RAM)**

- **32-bit MCU**
  1) **Arduino Uno R4**: An updated Uno with a **faster processor**, Built-in **WiFi**, more **memory**, and enhanced features. (**32 KB RAM**)
  2) **ESP32**: A **Wi-Fi and Bluetooth**-enabled microcontroller for **IoT projects**. (**520 KB**)
  3) **Pico**: A **small, low-cos**t, **high-performanc**e microcontroller great for **embedded development**. (**264 KB**)
  4) **Pico W**: A Pico with added **Wi-Fi connectivity** for IoT applications. (264 KB)
  5) **STM32**: A family of powerful 32-bit ARM Cortex-M microcontrollers for **demanding applications**. (**Up to 1 MB**)
  6) **Nano 33: A small board fea**turing a po**werfu**l 32-bit ARM Cortex-M **processor and sensors**. (**256 KB**)

Popular Single-Board Computer

- **Single-Board Computer (SBC) (low cost - OS - USB – HDMI- GPIO FOR SENSOR– 35-40)**
  1) **Raspberry Pi 5**: The latest Raspberry Pi model, offering a significant performance boost with a faster processor and improved features for **general-purpose computing and multimedia.**
  2) **Raspberry Pi Zero**: **A tiny, low-cost computer ideal** for embedded projects and situations where space is limited.
  3) **Jetson Nano**: A powerful single-board computer designed **for AI and machine learning applications**, featuring a dedicated GPU for accelerated processing.

Raspberry Pi 5

Raspberry Pi Zero

Jetson Nano

Microcontroller Units (MCUs) Vs. Single-Board Computers (SBCs)

Feature	Microcontroller Unit (MCU)	Single-Board Computer (SBC)
Definition	A compact integrated circuit designed for specific control tasks in embedded systems.	A fully functional computer on a single circuit board with a processor, memory, and storage.
Processing Power	Low-power, designed for simple tasks and real-time processing.	Higher processing power, capable of running operating systems like Linux or Windows.
Operating System	Typically runs firmware or bare-metal code (no OS).	Runs full operating systems like Linux, Raspberry Pi OS, or Windows IoT.
Memory & Storage	Limited RAM and no dedicated storage; uses flash memory.	Has RAM and expandable storage (e.g., microSD, eMMC, or SSD).
Power Consumption	Low power consumption, designed for battery-operated devices.	Higher power consumption, often requires external power adapters.
Connectivity	Limited interfaces (GPIO, UART, SPI, I2C); mostly used for embedded applications.	Supports USB, HDMI, Ethernet, Wi-Fi, and Bluetooth, enabling diverse applications.
Use Cases	Industrial automation, IoT devices, robotics, real-time control applications.	Edge computing, AI/ML applications, media centers, general computing tasks.
Examples	Arduino, ESP32, STM32, ATmega328.	Raspberry Pi, NVIDIA Jetson Nano.

- **MCUs** are ideal for real-time, low-power tasks in embedded systems.
- **SBCs** offer more computing power and flexibility for applications requiring an OS and multiple interfaces.

Sensor technology in
Autonomous Vehicle Systems

Sensor technologies are the eyes and ears of autonomous vehicles, converting physical phenomena into electrical signals and perceiving their environment.

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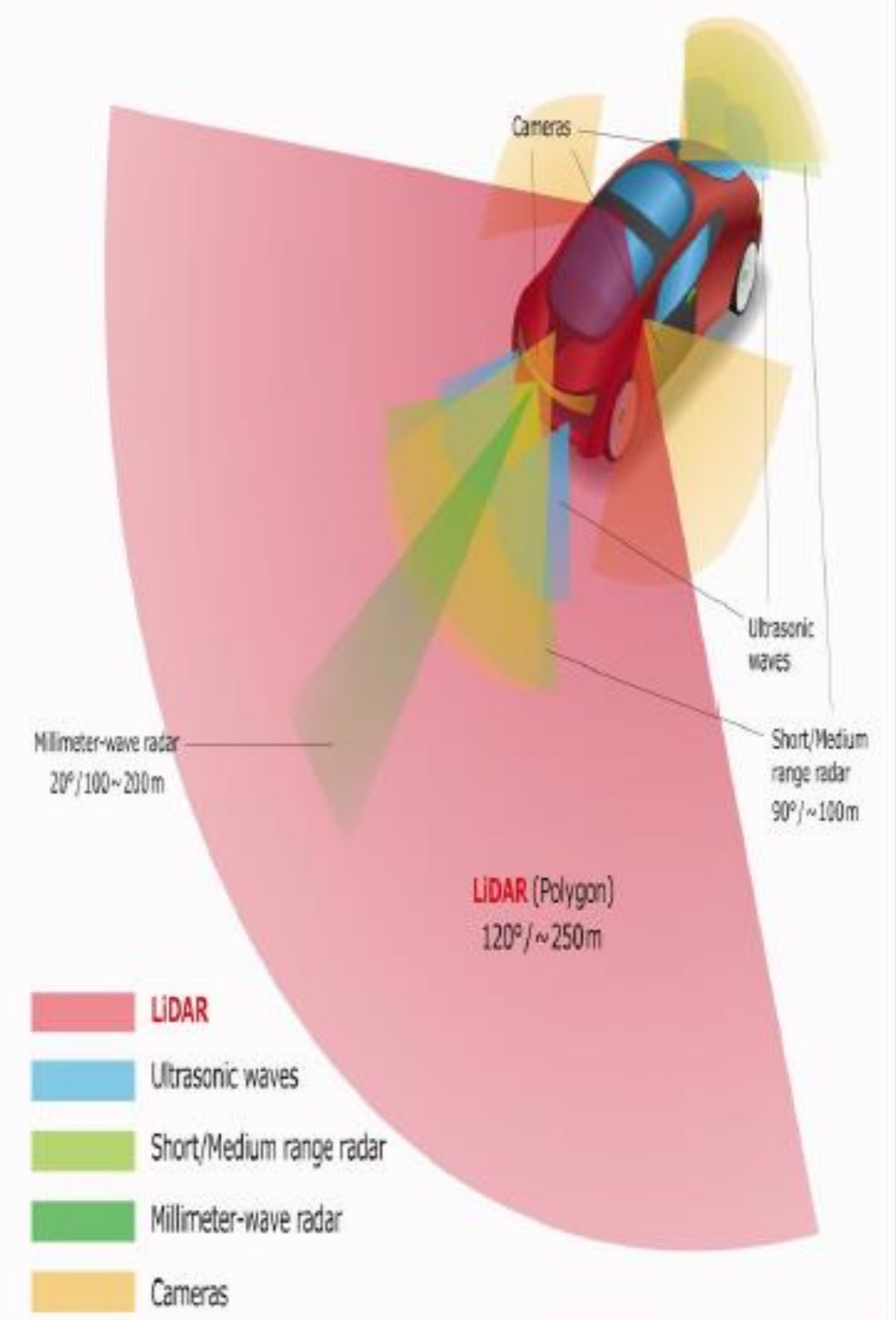
Types of Sensors

Autonomous vehicles rely on five main sensor types, each playing a crucial role in perception.



1. Camera Sensors

- Cameras rely on the principles of optics, focusing light through a lens onto an image sensor
  - Image sensors convert photons into electrical signals using photodiodes.

Purpose: Used for **object detection, lane tracking, sign recognition**, and identifying traffic signals.

Types of cameras:

- **RGB Cameras:** Provide high-resolution color images.
- **Infrared Cameras (IR):** Used for night vision and pedestrian detection.
- **Stereo Vision Cameras:** Two cameras placed at a distance to perceive depth (like human vision).

✓ Advantages:

- Provides rich visual information.
- Works well for lane and traffic sign recognition.

❑ Limitations:

- Performance degrades in low-light, fog, or glare.
- Lacks depth perception without stereo vision.

2. Radar Sensors

- Radar operates by transmitting electromagnetic waves and analyzing the reflected signals.
- The Doppler effect enables radar to measure the velocity of moving objects.
- Frequency-Modulated Continuous Wave (FMCW) radar provides high range resolution.

Purpose: Used for **distance measurement, object tracking, and speed detection**.

- **How it works:** Sends **radio waves** that bounce off objects, measuring the time it takes for waves to return.

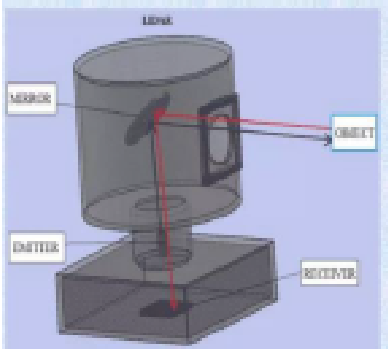
✓ **Advantages:**

- Works in **all weather conditions** (fog, rain, snow).
- Can detect **moving objects** like cars and pedestrians.
- Measures **velocity** of approaching vehicles.

❑ **Limitations:**

- Lower **spatial resolution** than LiDAR and cameras.
- Struggles with **small, static objects**.

3. LiDAR (Light Detection and Ranging)



- Lidar uses the time-of-flight principle, measuring the time taken for laser.
 pulses to travel to and from objects
- Lidar scanners employ rotating mirrors or solid-state beam steering to create 3D point clouds.
  - The wavelength of the laser affects its performance in different weather conditions

Purpose: Provides precise 3D mapping of surroundings.

How it works: Uses laser pulses to create a high-resolution 3D point cloud of the environment.

✓ Advantages:

- ✓ High accuracy in object detection and depth measurement.
- ✓ Works well for autonomous navigation and obstacle detection.
- ✓ Generates detailed 3D maps for localization.

❑ Limitations:

- ✓ Expensive and bulky.
- ✓ Performance is affected by fog, heavy rain, and dust.
- ✓ Requires high computational power for data processing.

4. Ultrasonic Sensors

•Ultrasonic sensors generate high-frequency sound waves and detect echoes to determine distance.

•The speed of sound in air is approximately 343 m/s at room temperature.

- **Purpose:** Used for short-range object detection, such as parking
assistance and low-speed maneuvering.

- **How it works:** Emits ultrasonic sound waves, measuring the reflection time.

- ✓ **Advantages:**
  - ✓ Works well in close-range scenarios (e.g., parking).
  - ✓ Inexpensive and widely used.

- ❑ **Limitations:**
  - ✓ Limited range (only a few meters).
  - ✓ Struggles with soft materials that absorb sound waves.

5. GPS & IMU (Inertial Measurement Unit)

- **GPS** relies on the concept of trilateration, calculating position based on the time difference of signals from multiple satellites.
- **IMU,** is a device that measures force, angular rate, and magnetic field of a body using three accelerometers, gyroscopes, and magnetometers. From the raw IMU data, linear velocity, altitude, and angular positions (relative to a global reference frame) of autonomous vehicles can be calculated.
- There are a few disadvantages of IMU due to how they work. The main disadvantage is that they only provide motion information and not location information. Another disadvantage is that the data can be affected by drift errors. These problems can be solved by implementing GPS devices.

Purpose: Provides vehicle positioning and motion tracking.

- GPS (Global Positioning System): Helps the vehicle determine its location.
- IMU (Inertial Measurement Unit): Measures acceleration, tilt, and rotation.

✓ **Advantages:**

- Crucial for navigation and localization.
- Helps in path planning.

❑ **Limitations:**

- GPS signals can be weak in tunnels or urban areas.
- IMU data drifts over time, requiring correction.

Conclusions

Sensors	Main roles / functions
Cameras	Recognition of shape and color, such as road signs and white lines [Lane departure prevention, Parking assist, Surround view]
Ultrasonic waves	Detection of objects at close distances [Parking Assist]
Short / Medium range radar	Detection of objects at close distances including relative speeds [Crossing vehicle warning, Rear collision warning]
Millimeter-wave radar	Detection of long-distance objects including relative velocity [ACC, LKAS, Simple pedestrian detection]
LiDAR	Detection of object position and distance / Azimuth resolution: High [Pedestrian detection, Emergency braking, Collision avoidance]